



Graphing Logarithmic Functions

Common logarithmic functions have base 10, while natural logarithmic functions have base e .

Definitions

If x is a positive real number, then

- $f(x) = \log x$ is a **common logarithmic function with base 10**.
- $g(x) = \ln x$ is a **natural logarithmic function with base e** .

Both of these logarithmic functions have domain $x > 0$ and a vertical asymptote at $x = 0$.

Logarithmic functions are related to exponential functions. In fact, a logarithmic function is always the inverse of some exponential function, so their graphs have similar shapes.

How to Graph a Logarithmic Function

Step 1 Find any x - or y -intercepts.

Step 2 Give the domain and the equation of the vertical asymptote.

Step 3 Make a table of values to find points on the graph.

Step 4 Draw a smooth curve using the vertical asymptote and points on the graph as a guide.

EXAMPLE 1

Graph $f(x) = \ln(x + 1)$.

SOLUTION

Step 1 Find any x - or y -intercepts.

To find the x -intercept, set $f(x) = 0$ and solve for x .

$$\begin{aligned}0 &= \ln(x + 2) \\ e^0 &= x + 2 \\ 1 &= x + 2 \\ -1 &= x\end{aligned}$$

To find the y -intercept, set $x = 0$ and solve for $f(x)$.

$$\begin{aligned}f(x) &= \ln(0 + 1) \\ &= \ln 1 \\ &= 0\end{aligned}$$

The x -intercept is $x = -1$, and the y -intercept is $y = 0$.

Step 2 Give the domain and the equation of the vertical asymptote.

Find the domain of $f(x)$ by solving the inequality $x + 1 > 0$.

$$\begin{aligned}x + 1 &> 0 \\ x &> -1\end{aligned}$$

Find the vertical asymptote by solving $x + 1 = 0$.

$$\begin{aligned}x + 1 &= 0 \\ x &= -1\end{aligned}$$

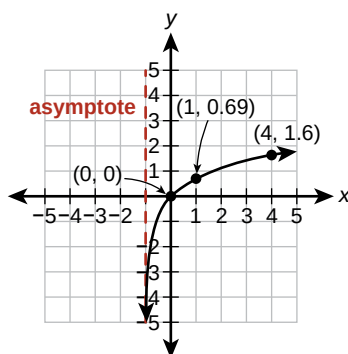
The domain is $x > -1$, and the vertical asymptote is the line $x = -1$.

► **Remember** You can only take the logarithm of a positive real number. The logarithm of 0 or a negative real number does not exist.

Step 3 Make a table of values to find points on the graph.

x	$f(x) = \ln(x + 1)$
0	$\ln(0 + 1) = \ln 1 = 0$
1	$\ln(1 + 1) = \ln 2 \approx 0.69$
4	$\ln(4 + 1) = \ln 5 \approx 1.6$

Step 4 Draw a smooth curve using the vertical asymptote and points on the graph as a guide.



► **Think About It** The graph never actually touches the vertical asymptote.

EXAMPLE 2

Graph $f(x) = 5 \log(2.3 - x)$.

SOLUTION

Step 1 Find any x - or y -intercepts.

To find the x -intercept, set $f(x) = 0$ and solve for x .

$$\begin{aligned} 0 &= 5 \log(2.3 - x) \\ 0 &= \log(2.3 - x) \\ 10^0 &= 2.3 - x \\ 1 &= 2.3 - x \\ x &= 1.3 \end{aligned}$$

To find the y -intercept, set $x = 0$ and solve for $f(x)$.

$$\begin{aligned} f(x) &= 5 \log(2.3 - 0) \\ &= 5 \log(2.3 - 0) \\ &= 5 \log(2.3) \\ &\approx 1.8 \end{aligned}$$

The x -intercept is $x = 1.3$, and the y -intercept is $y \approx 1.8$.

Step 2 Give the domain and the equation of the vertical asymptote.

Find the domain of $f(x)$ by solving the inequality $2.3 - x > 0$.

$$\begin{aligned} 2.3 - x &> 0 \\ -x &> -2.3 \\ x &< 2.3 \end{aligned}$$

Find the vertical asymptote by solving $2.3 - x = 0$.

$$\begin{aligned} 2.3 - x &= 0 \\ -x &= -2.3 \\ x &= 2.3 \end{aligned}$$

The domain is $x < 2.3$, and the vertical asymptote is the line $x = 2.3$.

Step 3 Make a table of values to find points on the graph.

x	$f(x) = 5 \log(2.3 - x)$
-3	$5 \log(2.3 - (-3)) = 5 \log 5.3 \approx 3.6$
-1	$5 \log(2.3 - (-1)) = 5 \log 3.3 \approx 2.6$
2	$5 \log(2.3 - 2) = 5 \log 0.3 \approx -2.6$

Step 4 Draw a smooth curve using the vertical asymptote and points on the graph as a guide.

